

AEGIS

Child Safety Band

FYP / II — Meeting Records

Meeting Forms 11 – 20

with Supervisor Remarks & Progress Notes

Supervisor: Dr. Altaf Hussain

Members: Barira Sarfraz | Nabeeha Zahid

Reg. No.: 222201044 | 222201009

21 April 2026 – 18 June 2026

Meeting Form No. 11**Date: 21 April, 2026 Time: 10:30 AM**

Project Title	AEGIS — Child Safety Band	Supervisor	Dr. Altaf Hussain
Members	Barira Sarfraz Nabeeha Zahid	Reg. No.	222201044 222201009

Action Items

1. Get Firebase connected to the Flutter app — Firestore and Auth set up.
2. Agree on a folder structure and clean up the pubspec dependencies.
3. Sketch out a rough plan for what we are building across the app, firmware, and ML.

Progress on Previous Items (Supervisor Remarks — based on Form 10 action items)

- Diagrams refined. Sequence diagram updated with BLE flow.
- Use-case diagram aligned with defended concept.
- Design documentation complete. Ready for implementation phase.

Meeting Form No. 12**Date: 28 April, 2026 Time: 11:00 AM**

Project Title	AEGIS — Child Safety Band	Supervisor	Dr. Altaf Hussain
Members	Barira Sarfraz Nabeeha Zahid	Reg. No.	222201044 222201009

Action Items

1. Set up the bottom nav bar and make sure routing works across all five screens.
2. Build out basic page layouts for Home, Map, Alerts, Profile, and Settings.
3. Go over the screens and check they match the Velvet Night design we have been following.

Progress on Previous Items (Supervisor Remarks — based on Form 11 action items)

- Firebase connected. Firestore and Auth initialised.
- Folder structure and dependencies finalised.
- Development roadmap drafted and approved.

Meeting Form No. 13**Date: 07 May, 2026 Time: 10:00 AM**

Project Title	AEGIS — Child Safety Band	Supervisor	Dr. Altaf Hussain
Members	Barira Sarfraz Nabeeha Zahid	Reg. No.	222201044 222201009

Action Items

1. Get email and password login and signup working with Firebase Auth.
2. Add field validation and keep the user logged in after closing the app.
3. Test the full sign-in and sign-out flow on the emulator and a real phone.

Progress on Previous Items (Supervisor Remarks — based on Form 12 action items)

- Bottom nav bar implemented. Routing working across all screens.
- Page layouts built with consistent styling.
- Screens match Velvet Night design. Spacing corrected.

Meeting Form No. 14**Date: 13 May, 2026 Time: 11:30 AM**

Project Title	AEGIS — Child Safety Band	Supervisor	Dr. Altaf Hussain
Members	Barira Sarfraz Nabeeha Zahid	Reg. No.	222201044 222201009

Action Items

1. Add Google Maps and show the child's live position on the tracking screen.
2. Set up a Firestore listener so the location updates automatically.
3. Put a proper marker and small avatar on the map so the child is easy to spot.

Progress on Previous Items (Supervisor Remarks — based on Form 13 action items)

- Login and signup working with Firebase Auth.
- Form validation and session persistence in place.
- Tested on emulator and physical device. All passing.

Meeting Form No. 15**Date: 20 May, 2026 Time: 10:30 AM**

Project Title	AEGIS — Child Safety Band	Supervisor	Dr. Altaf Hussain
Members	Barira Sarfraz Nabeeha Zahid	Reg. No.	222201044 222201009

Action Items

1. Let the parent draw a safe zone circle on the map and adjust its size.
2. Save the zone to Firestore so it loads back next time the app opens.
3. Fire a notification when the child steps outside the safe zone.

Progress on Previous Items (Supervisor Remarks — based on Form 14 action items)

- Google Maps integrated. Live location displaying correctly.
- Firestore listener active. Coordinates updating in real time.
- Custom marker and avatar added to tracking screen.

Meeting Form No. 16**Date: 26 May, 2026 Time: 11:00 AM**

Project Title	AEGIS — Child Safety Band	Supervisor	Dr. Altaf Hussain
Members	Barira Sarfraz Nabeeha Zahid	Reg. No.	222201044 222201009

Action Items

1. Write firmware to read GPS data from the Neo-6M and send it over BLE.
2. Broadcast the device ID and coordinates from the ESP32 to the paired phone.
3. Test it on the real hardware and confirm the Flutter app receives the data correctly.

Progress on Previous Items (Supervisor Remarks — based on Form 15 action items)

- Safe zone circle implemented. Draggable and resizable.
- Zone data saved to Firestore. Loads correctly on relaunch.
- Notification triggers on safe zone breach. Tested successfully.

Meeting Form No. 17**Date: 03 June, 2026 Time: 10:30 AM**

Project Title	AEGIS — Child Safety Band	Supervisor	Dr. Altaf Hussain
Members	Barira Sarfraz Nabeeha Zahid	Reg. No.	222201044 222201009

Action Items

1. Load the TFLite activity model onto the ESP32 and run it on MPU-6050 sensor data.
2. Normalise the accelerometer readings before passing them to the model.
3. Verify the model is picking up idle, walking, running, and fall correctly.

Progress on Previous Items (Supervisor Remarks — based on Form 16 action items)

- ESP32 reading GPS from Neo-6M. NMEA parsing working.
- BLE broadcasting device ID and GPS payload. Verified.
- Flutter app receiving and parsing BLE data correctly.

Meeting Form No. 18**Date: 09 June, 2026 Time: 11:00 AM**

Project Title	AEGIS — Child Safety Band	Supervisor	Dr. Altaf Hussain
Members	Barira Sarfraz Nabeeha Zahid	Reg. No.	222201044 222201009

Action Items

1. Set up FCM so parents get push alerts when something happens.
2. Make sure all three alerts fire — safe zone exit, fall detected, and low battery.
3. Add an alert history screen in the app showing what happened and when.

Progress on Previous Items (Supervisor Remarks — based on Form 17 action items)

- TFLite model running on ESP32. Inference on MPU-6050 confirmed.
- Preprocessing pipeline in place. Readings normalised before inference.
- Activity labels and fall detection at acceptable accuracy.

Meeting Form No. 19**Date: 16 June, 2026 Time: 10:00 AM**

Project Title	AEGIS — Child Safety Band	Supervisor	Dr. Altaf Hussain
Members	Barira Sarfraz Nabeeha Zahid	Reg. No.	222201044 222201009

Action Items

1. Run a full end-to-end test on the BLE link between the band and the app.
2. Stress-test the Firestore sync to see how it holds up with live location data.
3. Write down any bugs we find and plan fixes for the next session.

Progress on Previous Items (Supervisor Remarks — based on Form 18 action items)

- FCM set up. Push notifications reaching parent device.
- All three alert types triggering correctly in tests.
- Alert history screen complete. Entries showing with timestamps.

Meeting Form No. 20**Date: 18 June, 2026 Time: 11:30 AM**

Project Title	AEGIS — Child Safety Band	Supervisor	Dr. Altaf Hussain
Members	Barira Sarfraz Nabeeha Zahid	Reg. No.	222201044 222201009

Action Items

1. Walk through the whole system — hardware, firmware, and the app — and review everything.
2. Finalise the demo script and any remaining documentation for FYP submission.
3. Confirm everything is in order and the project is ready for the final evaluation.

Progress on Previous Items (Supervisor Remarks — based on Form 19 action items)

- Full BLE pipeline tested. Packet-loss bug found and fixed.
- Firestore sync stable under load. Response times acceptable.
- All bugs resolved. System ready for final evaluation.